

Computer Networking

Chapter 2: Application Layer

Questions & Answers

Question 1

Performance Analysis

Consider a scenario:

- Average object size: 1 Mbit
- Request rate: 15 requests/sec
- LAN bandwidth: 100 Mbps
- Access link bandwidth: 15 Mbps
- Internet delay: ≈ 2 seconds

Answer the following:

1. Compute the traffic intensity on the LAN and on the access link.
2. Explain why response time without caching is unacceptable.
3. If the access link is upgraded to 100 Mbps, what will be the response time?
4. With web caching (hit rate = 0.4), compute the average delay.

Answer

(a) Traffic Intensity

$$\text{LAN intensity} = \frac{15 \times 1}{100} = 0.15$$

$$\text{Access link intensity} = \frac{15 \times 1}{15} = 1.0$$

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(b) Without Caching - LAN delay $\approx$ negligible. - Access link delay $\rightarrow$ extremely high as utilization $\rightarrow 1$. - Response time $\approx$ minutes $\Rightarrow$ unacceptable.
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(c) Upgrading Access Link to 100 Mbps

$$\text{Traffic Intensity} = \frac{15 \times 1}{100} = 0.15$$

- Internet delay dominates $\Rightarrow$ response time ≈ 2 s. - But costly solution.

(d) With Web Caching (hit rate = 0.4)

$$\begin{aligned}\text{Average Delay} &= (0.4)(0.01 \text{ s}) + (0.6)(2.01 \text{ s}) \\ &= 0.004 + 1.206 \approx 1.21 \text{ s}\end{aligned}$$

Final Answer: Web caching reduces the response time to about **1.2** s.

Question 2

Write your next question here.

Answer

Write the solution here.